PharmAchieve

PRINCIPLES OF INFECTIOUS DISEASE & ANTIMICROBIAL THERAPY I

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care: Clinical Care

Providing Care: Distribution

Knowledge & Expertise

Communication & Collaboration

Leadership & Stewardship

Professionalism

LEGEND

Abx	Antibiotic
ADR	Adverse Drug Reactions
AMG	Aminoglycosides
Amox/Clav	Amoxicillin/Clavulanic Acid
Amp	Ampicillin
AUC	Area Under the Curve
β-lactams	Beta-lactams
C/I	Contraindications
Carbapen	Carbapenem
Ceph	Cephalosporins
Cipro	Ciprofloxacin
Clinda	Clindamycin
Dapto	Daptomycin
Enterobact	Enterobacter
Esp.	Especially
FQ	Fluoroquinolones
G+ve/G-ve	Gram Positive/Negative
Gen	Generation
Group A	Group A Streptococcus
HIV	Human Immunodeficiency Virus
Ifx	Infection
IV	Intravenous
Kleb	Klebsiella
Linez	Linezolid
Metro	Metronidazole
MI	Myocardial Infarction
MIC	Minimum Inhibitory Concentration
Min	Minutes
MRSA	Methicillin-Resistant
MSSA	Methicillin-Susceptible
NPO	Nil by mouth
PAE	Post-antibiotic Effect
PD	Pharmacodynamics
Pen	Penicillin
Pip/Taz	Piperacillin/Tazobactam
PK	Pharmacokinetics
PO	By Mouth
Rxn	Reaction
S.pneumo	Streptococcus Pneumoniae
SA	Staphylococcus Aureus
SLE	Systemic Lupus Erythematosus
SMX/TMP	Sulfamethoxazole/Trimetho-prim
Spp	Species
TB	Tuberculosis
Tetra	Tetracycline
Vanco	Vancomycin
VRE	Vancomycin-Resistant Enterococcus
WBC	White Blood Cells

PRINCIPLES OF INFECTION

Case 1

A 35-year-old woman (62 kg, 162 cm) presents to your urgent care clinic with 2 days of fever (39.3 °C), chills, right flank pain, and dysuria. She has nausea and poor oral intake.

PMH: Crohn’s disease (on infliximab q8 weeks), mild asthma, anxiety.
Medications: infliximab, budesonide inhaler, sertraline, oral contraceptive.
Allergies: none known.
Social history: school teacher, returned from a 3-week trip to India 3 weeks ago, where she ate street food and was hospitalized briefly for traveler’s diarrhea. She self-started ciprofloxacin 500 mg BID × 2 days from an old prescription before presentation.
Urine dipstick: positive nitrite and leukocyte esterase.
Labs: WBC 16 × 10^9/L, Cr 100 µmol/L (baseline 70), ALT 38 U/L, lactate 1.4 mmol/L.
Urine microscopy: > 50 WBC/hpf, no yeast.
CT abdomen/pelvis: mild perinephric stranding, no obstruction or stones.

PRINCIPLES OF INFECTION

PATHOGENICITY: ability to *cause disease* in host organism

PATHOGENESIS: *process* in which a disease develops by various *portals* of entry

VIRULENCE: extent or *degree of infection*

Virulence factors enhance pathogenicity:

- Capsules
- Cell wall components
- Enzymes

MICROBIOME

- Microorganisms living on and inside of human body:
  - *Generally NOT harmful*
  - Potentially *beneficial* by:
    - Preventing overgrowth of pathogenic microbes
    - Producing useful substances (e.g. vitamins)

Host Response to Infection

Innate Immunity

- Defense at birth
- Rapid response
- NO memory response

+

Adaptive Immunity

- Specific
- Changes over time
- Slower than innate
- Memory response

INFECTION SIGNS & SYMPTOMS

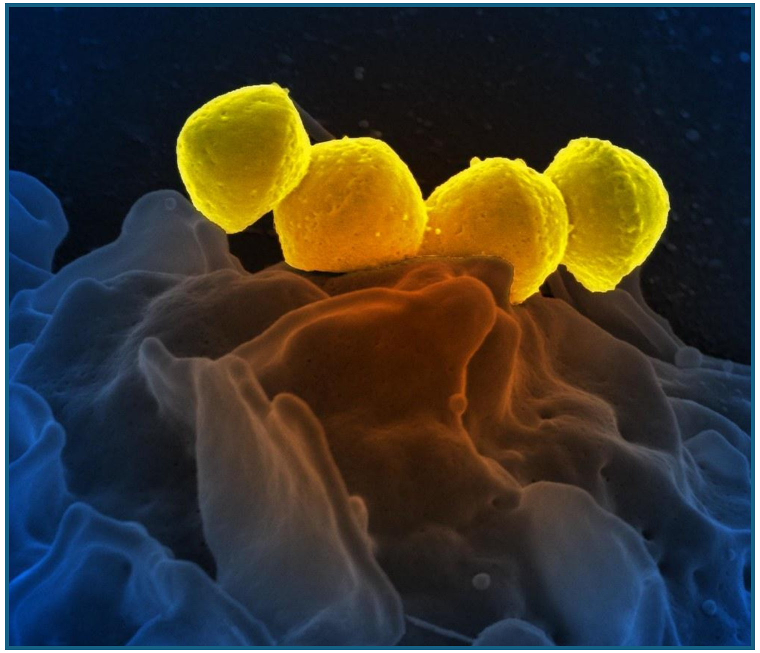
Local Signs/Symptoms of Infection

- Pain
- Inflammation
- Redness
- Cough
- Rhinorrhea
- Sore throat

Systemic Signs/Symptoms of Infection

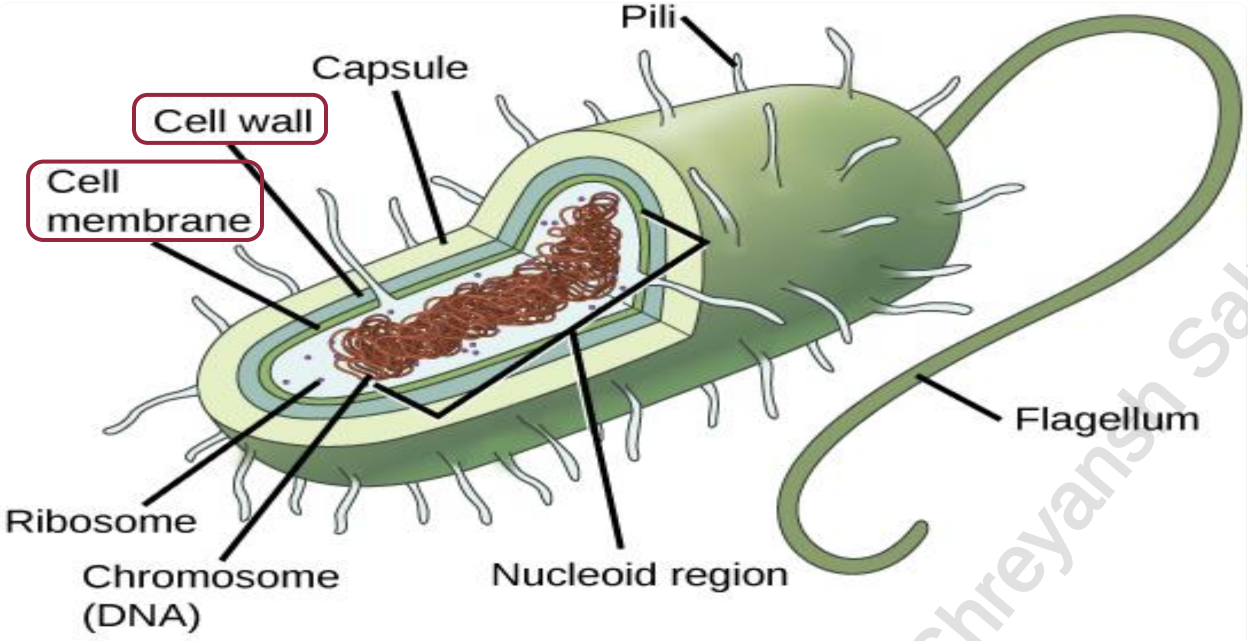
- Fever
- Malaise
- Leukocytosis
- Organ dysfunction
- Hypotension/Tachycardia

WHAT CAUSES INFECTION?



- Bacteria
- Viruses
- Parasites
- Fungi
- Other

CLINICAL MICROBIOLOGY REVIEW: BACTERIA



Cell wall: provides structural and homeostatic support

Plasma membrane: Thin structure consisting of ***phospholipids, carbohydrates, and proteins***

GRAM-POSITIVE

- Stains purple
- Stain binds peptidoglycan cell wall

GRAM-NEGATIVE

- Stains pink (i.e. doesn't stain)
- Has an outer plasma membrane

COMMON BACTERIA OF MEDICAL IMPORTANCE

GRAM POSITIVE

- Staphylococci
  - *S. aureus*
  - *S. epidermitis* and other coagulase negative staphylococci
- Streptococci
  - *Streptococcus pneumoniae*
  - Viridans group streptococci
  - Beta-hemolytic streptococci (Group A, B, C, G, etc)
- Enterococci
  - *E. faecalis*
  - *E. faecium*
- *Listeria monocytogenes*

GRAM NEGATIVE

- Enterobacteriales (i.e. gut bacteria)
  - *Proteus spp, E.coli and Klebsiella pneumoniae (PEcK)*
  - *Citrobacter freundii, Enterobacter spp., K. aerogenes*
- Respiratory tract gram negatives
  - *Haemophilus influenzae*
  - *Neisseria meningitidis*
  - *Moraxella catarrhalis*
- Lactose non-fermenting gram negatives
  - *Pseudomonas aeruginosa*
  - *Stenotrophomonas maltophilia*

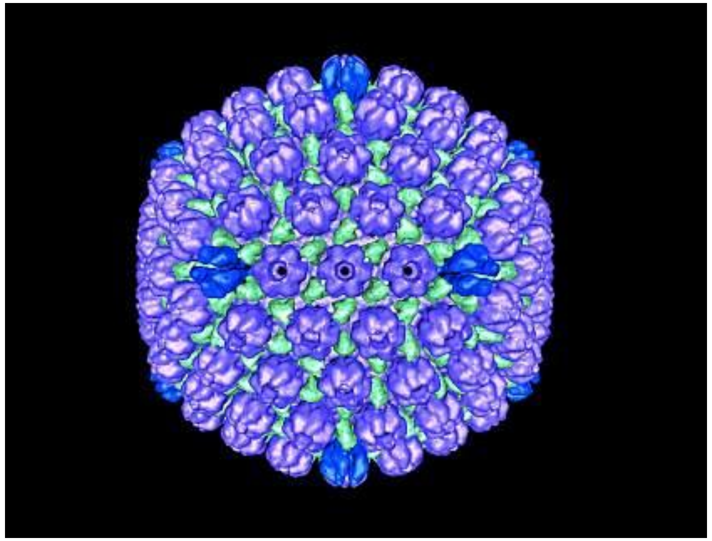
ATYPICAL

- *Legionella spp.*
- *Mycoplasma spp.*
- *Chlamydia/Chlamydophila spp.*

ANAEROBES

- Oral anaerobes: Peptostreptococci, *Fusobacterium spp., Prevotella spp.*
- Gut anaerobes: *Bacteroides spp, Clostridiodes spp.*

CLINICAL MICROBIOLOGY REVIEW: VIRUSES



- *RNA*
- *DNA*
- *Retroviruses*

Mild Disease

- The common cold
- Influenza
- Varicella
- Herpes Simplex Virus

Severe Disease

- Influenza
- Varicella
- Herpes Simplex Virus
- Ebola

CLINICAL MICROBIOLOGY REVIEW: FUNGI

- *Yeasts*
- *Molds*
- *Dimorphic Fungi*

Mild Disease

- Vulvovaginal Candidiasis
- Dandruff
- Oral Candidiasis (Thrush)
- Athlete's Foot

Severe Disease

- Candidemia/Endocarditis
- Invasive Aspergillosis
- Mucormycosis
- Invasive blastomycosis

CLINICAL MICROBIOLOGY REVIEW: PARASITES

- *Ectoparasites*
- *Round worms*
- *Protozoa*
- *Amoeba*

Mild Disease

- Bed bugs
- Scabies
- Pinworms
- Giardiasis

Severe Disease

- Malaria
- Echinococcus
- Schistosomiasis
- Strongyloidiasis

SELECTING ANTIMICROBIALS

FACTORS INVOLVED IN EMPIRIC THERAPY SELECTION

1. Patient risk/severity of illness
2. Patient specific factors (allergies, pregnancy, renal/hepatic impairment)

3. Infectious syndrome (e.g. superficial vs. deep skin infection)
4. Likely microbiology (e.g. site of infection, patient history and exposures)
5. Likely antimicrobial susceptibility (e.g. antibiogram; see next slide)

6. Antimicrobial bioavailability/penetration (e.g. can drug get to the site of infection?)
7. Antimicrobial efficacy and safety
8. Antimicrobial dosing and route

Host Factors
Organism Factors
Drug Factors

PATIENT RISK/SEVERITY OF ILLNESS

Host Factors

Risk	Severity of Illness
• Immune system status • Age • Co-morbid conditions	• Vital signs • Acute organ dysfunction • Other signs/symptoms

PATIENT SPECIFIC FACTORS

Host Factors

Factor	Comments
Allergies	Type 1-4 hypersensitivity reactions
Renal Function	Acute / Chronic Renal Dysfunction impacting drug clearance Risk of nephrotoxic agents impacting pre-existent renal dysfunction
Hepatic Function	Acute / Chronic Hepatic Dysfunction Impacting drug clearance Risk of hepatotoxic agents impacting pre-existent hepatic dysfunction
Body Mass	Potential impact on volume of distribution
Drug-drug interactions	Risk of antimicrobial drug interaction with current medications (e.g. linezolid with SSRI)
Drug-disease interactions	Risk of antimicrobial interaction with current disease (e.g. macrolide antimicrobials and myasthenia gravis)
Route access	Oral, intravenous availability (i.e. NPO status, no available IV lines, patient refusal)
Adherence	Willingness or ability to self-administer medication

INFECTIOUS SYNDROME

Organism Factors

Site of Infection

- Central nervous system (meningitis, encephalitis)
- Lung (pneumonia, pleural space infection
- Heart (infective endocarditis)
- Abdomen (appendicitis, peritonitis)
- Genitourinary tract (cystitis, pyelonephritis)
- Bone and Joint (osteomyelitis, prosthetic joint infection)
- Skin and Soft Tissue (cellulitis, necrotizing fasciitis)
- Primary bloodstream infection (*S. aureus, Candida spp.*)

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Urine dipstick: positive nitrite and leukocyte esterase.
Labs: WBC 16 × 10^9/L, Cr 100 μmol/L (baseline 70), ALT 38 U/L, lactate 1.4 mmol/L.
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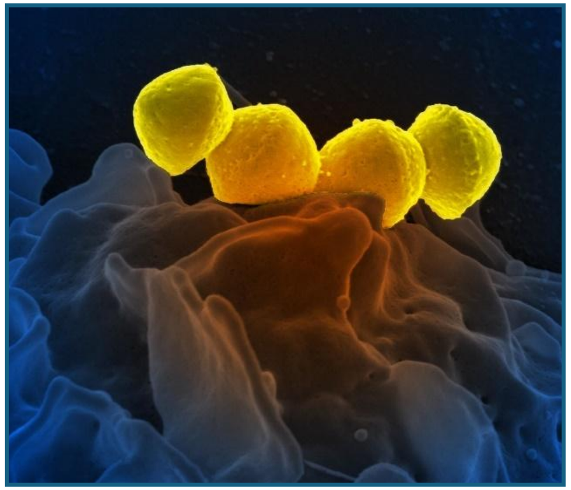
LIKELY MICROBIOLOGY

Syndrome

Pathogenesis of Infection (i.e., how did it happen?)

- Host factors
- Pathogen factors
- Environmental factors

Organism Factors



- Bacteria
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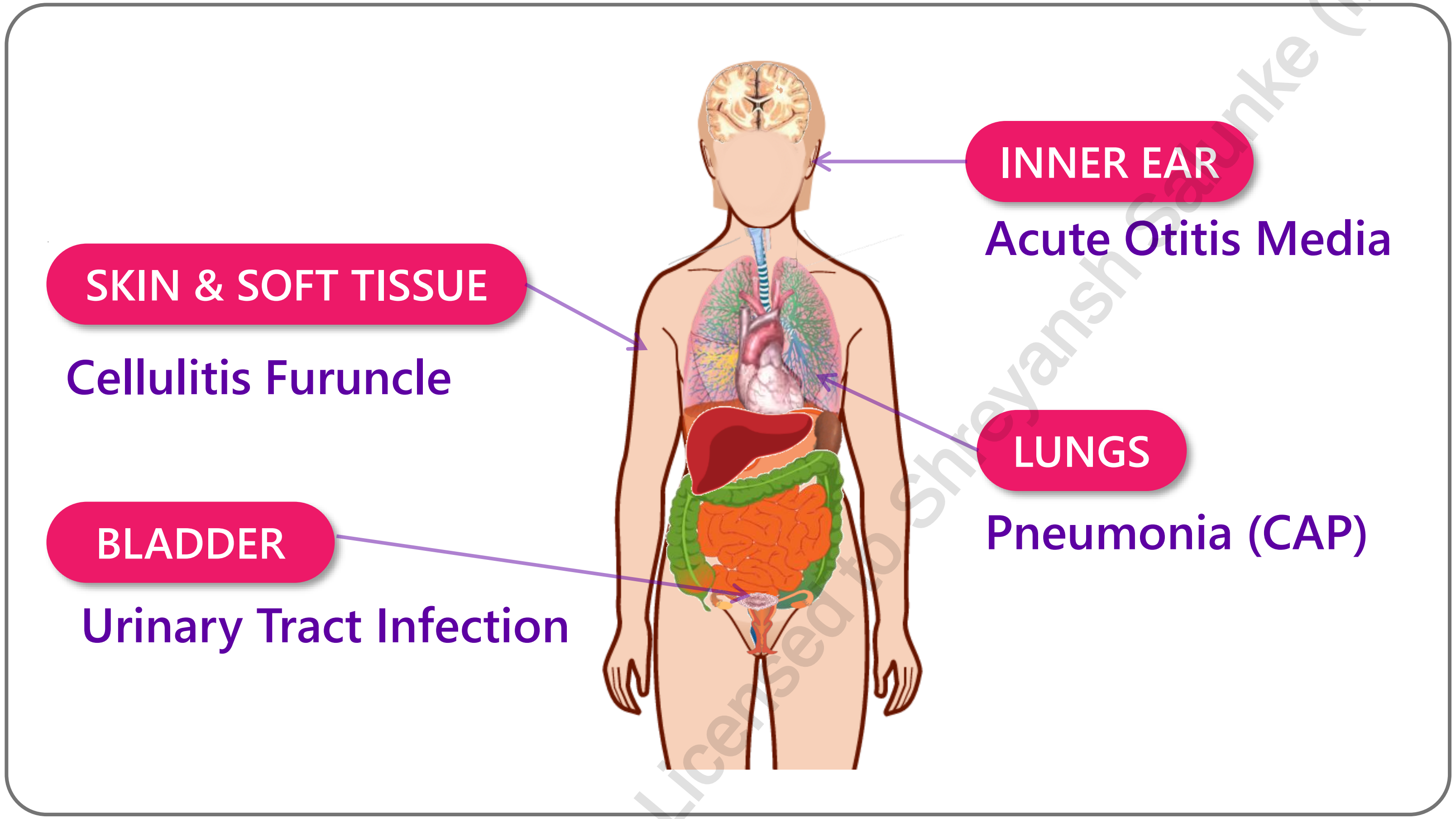
ID OVERVIEW

INTERACTIVE INFECTIOUS DISEASE DIAGRAM

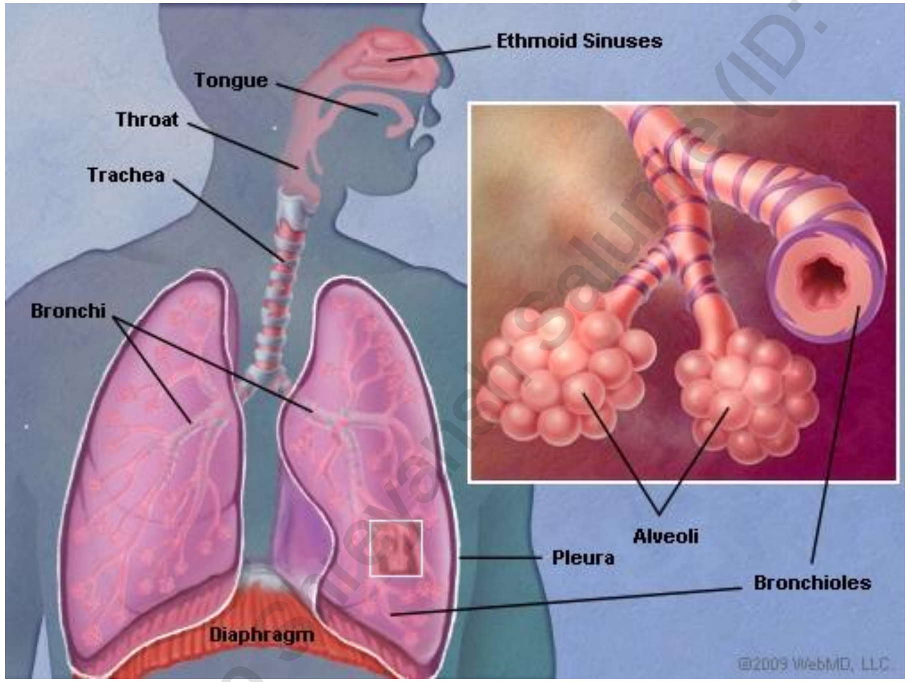
Organism Factors



COMMUNITY ACQUIRED PNEUMONIA

COMMON BUGS

- *S. pneumoniae*
- Respiratory viruses
- *M. pneumoniae*
- *C. pneumoniae*
- *H. influenzae*

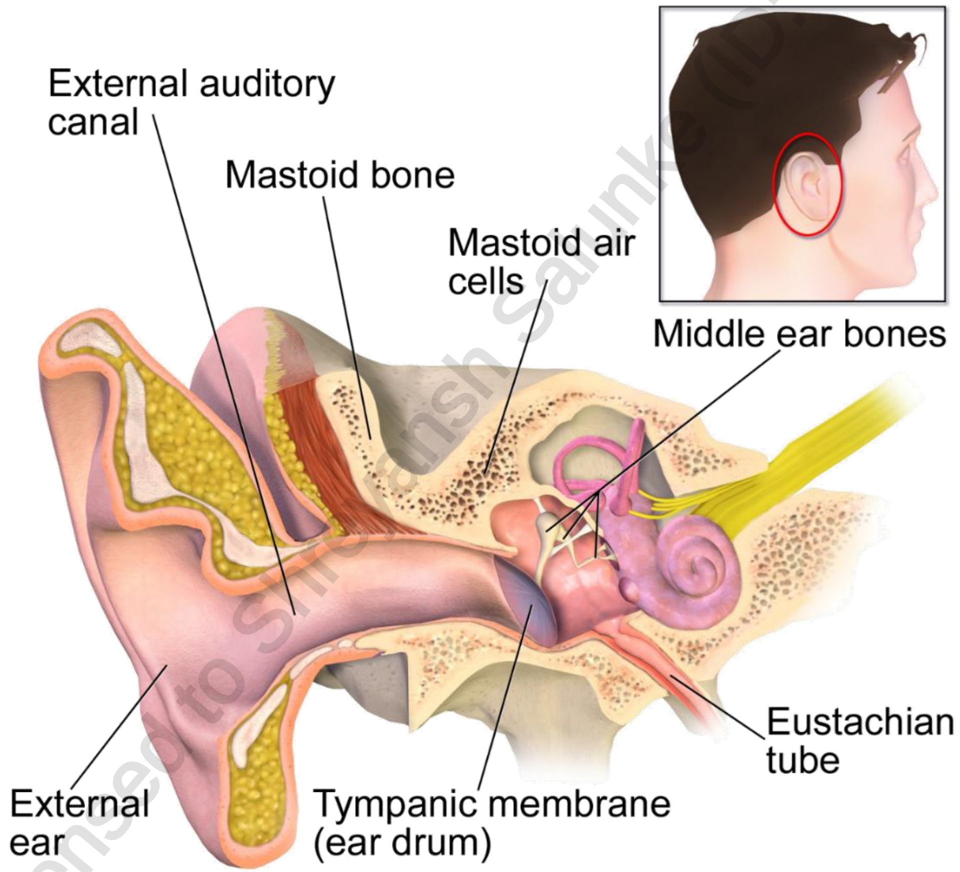


Organism Factors

ACUTE OTITIS MEDIA

COMMON BUGS

- *S. pneumoniae*
- *H. influenzae*
- *M. catarrhalis*



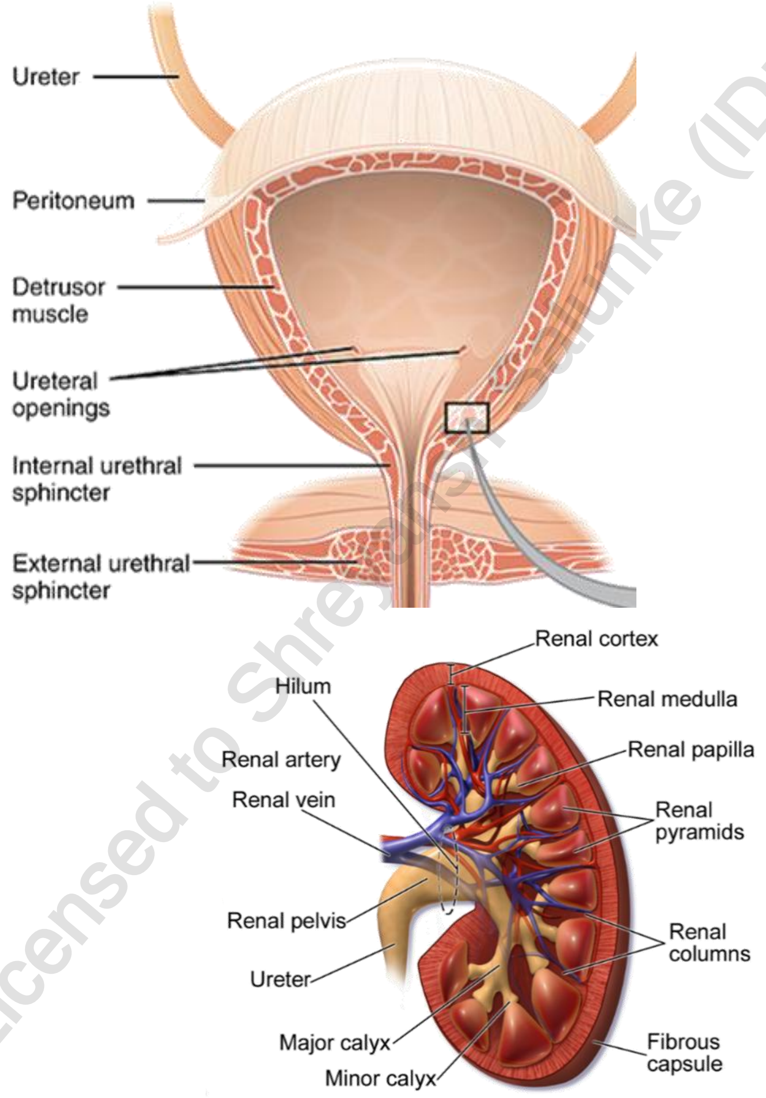
Ear Anatomy and Mastoid Bone

Organism Factors

UTI

COMMON BUGS

- *E. coli*
- *Proteus spp.*
- *S. saprophyticus*



Organism Factors

SKIN & SOFT TISSUE INFECTIONS

Organism Factors

FURUNCLES/CARBUNCLES

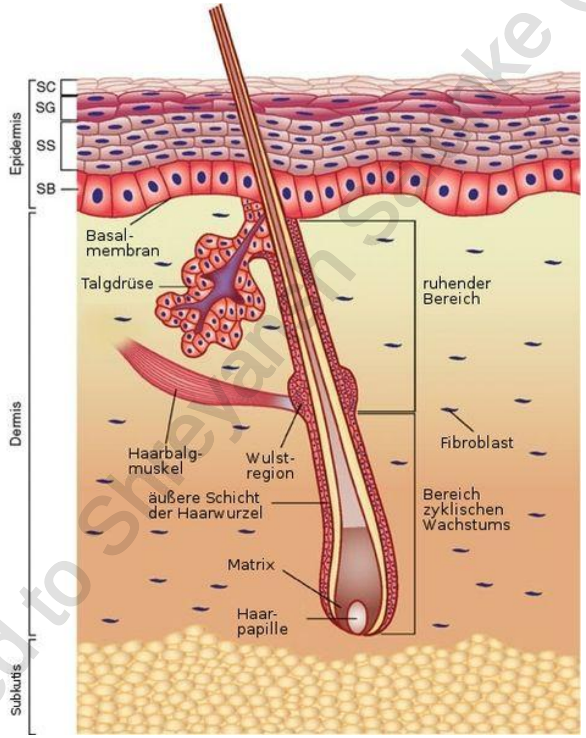
COMMON BUGS

- *S. aureus (MSSA, MRSA)*

CELLULITIS

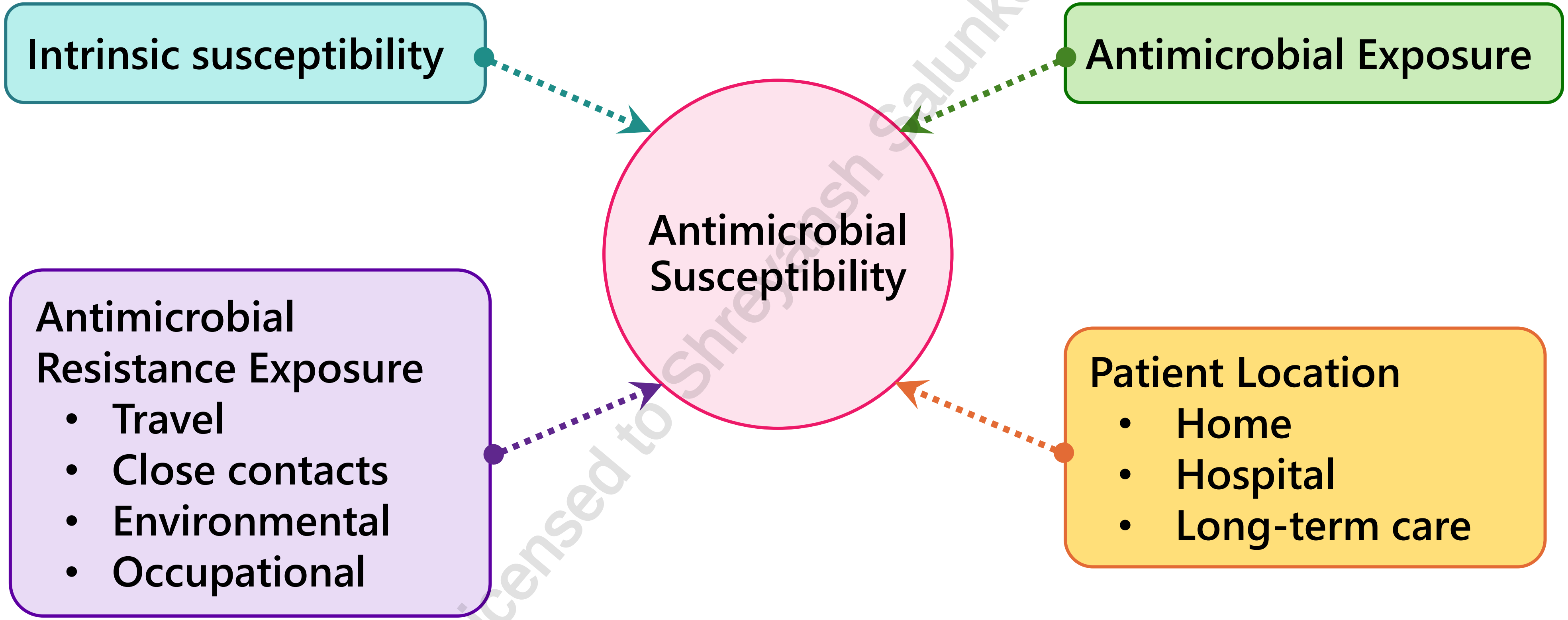
COMMON BUGS

- *Streptococcal spp*



LIKELY ANTIMICROBIAL SUSCEPTIBILITY

Organism Factors



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ANTIMICROBIAL BIOAVAILABILITY, PENETRATION AND TARGET SITE CONCENTRATION

Drug Factors

Bioavailability (F) → Tissue Penetration → Target Site Concentration

AVAILABLE ANTIMICROBIAL AGENTS AND MECHANISMS OF ACTION

Drug Factors

DNA Synthesis Inhibitors

- Fluoroquinolones
- Nitroimidazoles (ex. Metronidazole)

Cell Wall Inhibitors

- Penicillins
- Cephalosporins
- Carbapenems,
- Glycopeptides (ex. vancomycin)
- Beta Lactam-Beta Lactamase Inhibitors
- Lipopeptides (ex. daptomycin)
- Phosphonics (ex. Fosfomycin)

Protein Synthesis Inhibitors

- Macrolides (ex. azithromycin)
- Aminoglycosides (ex. tobramycin)
- Tetracyclines/Glycylcyclines (ex. doxycycline/tigecycline)
- Lincosamides (ex. clindamycin)
- Oxazolidinones (ex. linezolid)

Folate Synthesis Inhibitors

- Sulfamethoxazole-Trimethoprim

Miscellaneous

- Nitrofurantoin

ANTIMICROBIAL "EFFICACY" AND SAFETY

Drug Factors

Efficacy	
BACTERICIDAL ABX	BACTERIOSTATIC ABX
β-lactams	Tetracyclines
Vancomycin	Macrolides
Aminoglycosides	Clindamycin
Daptomycin	Linezolid
Fluoroquinolones	

Safety (i.e. risk of ADR)		
Low risk	Moderate risk	High risk
• β lactams	• Macrolides • Fluoroquinolones	• Chloramphenicol • Aminoglycosides

ANTIMICROBIAL PK/PD TARGETS

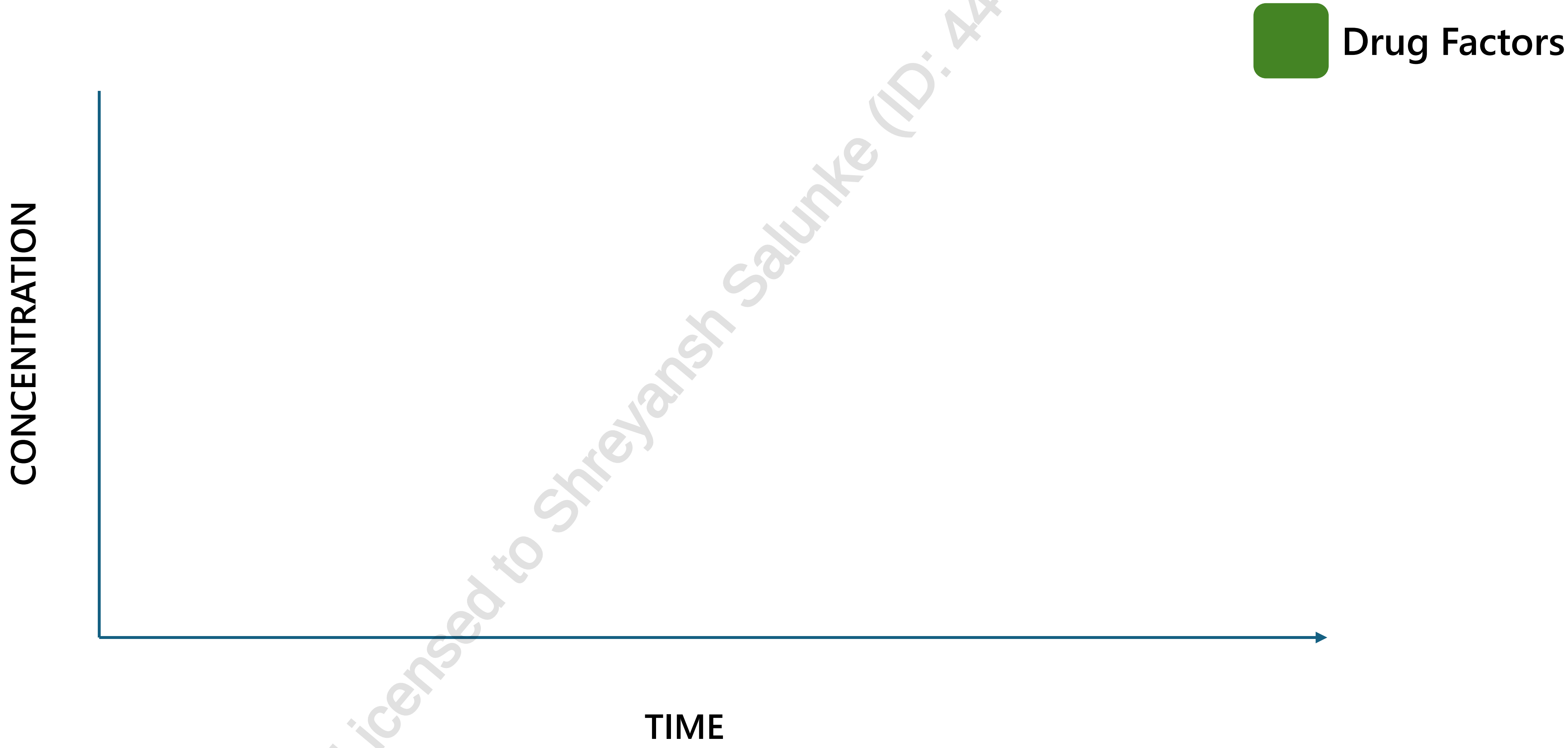
Drug Factors

Time Dependent	
Drug	Target
β-lactams	T>MIC

Concentration:Time Dependent	
Drug	Target
Vancomycin	AUC:MIC

Concentration Dependent	
Drug	Target
AG, FQs	C>MIC

ANTIMICROBIAL PK/PD TARGETS



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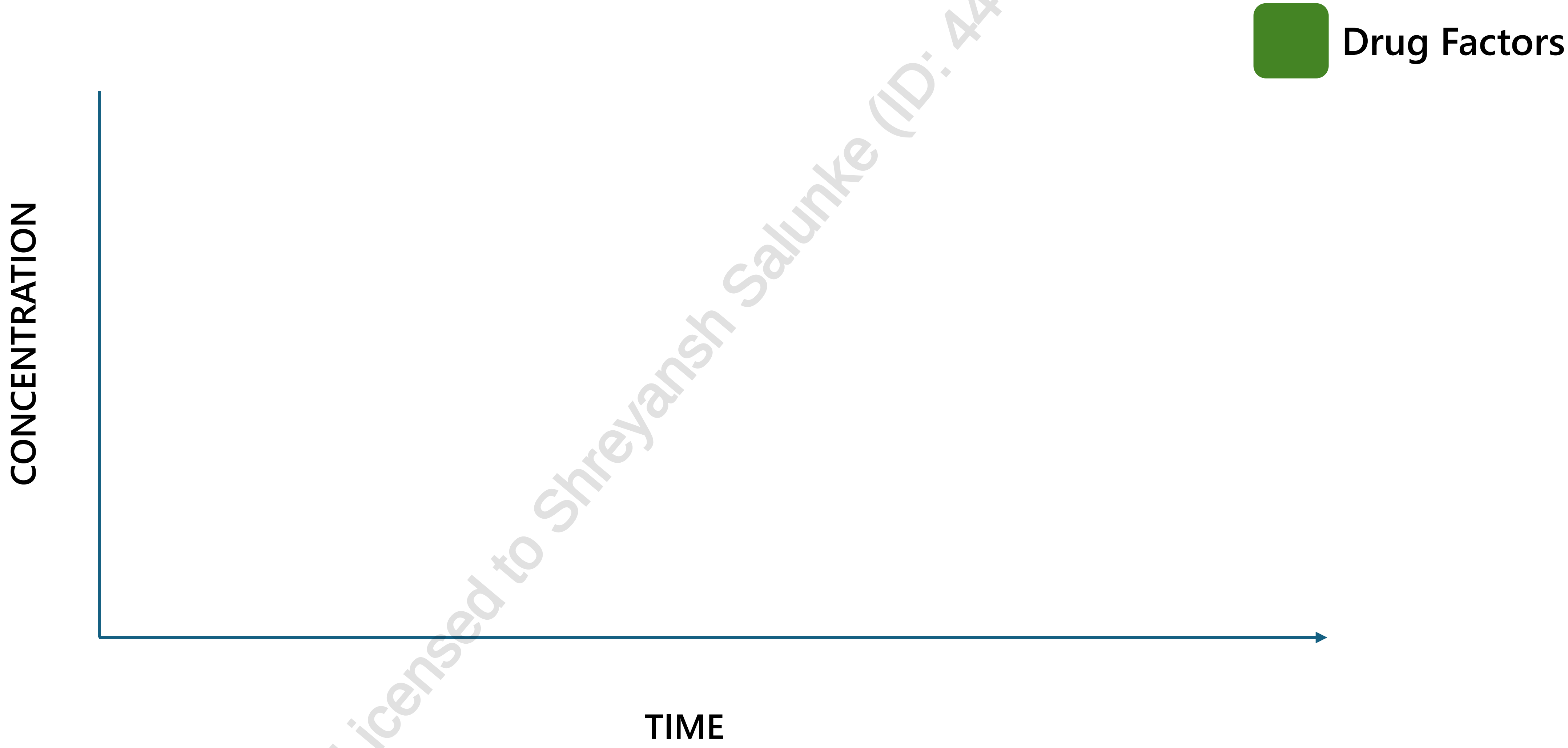

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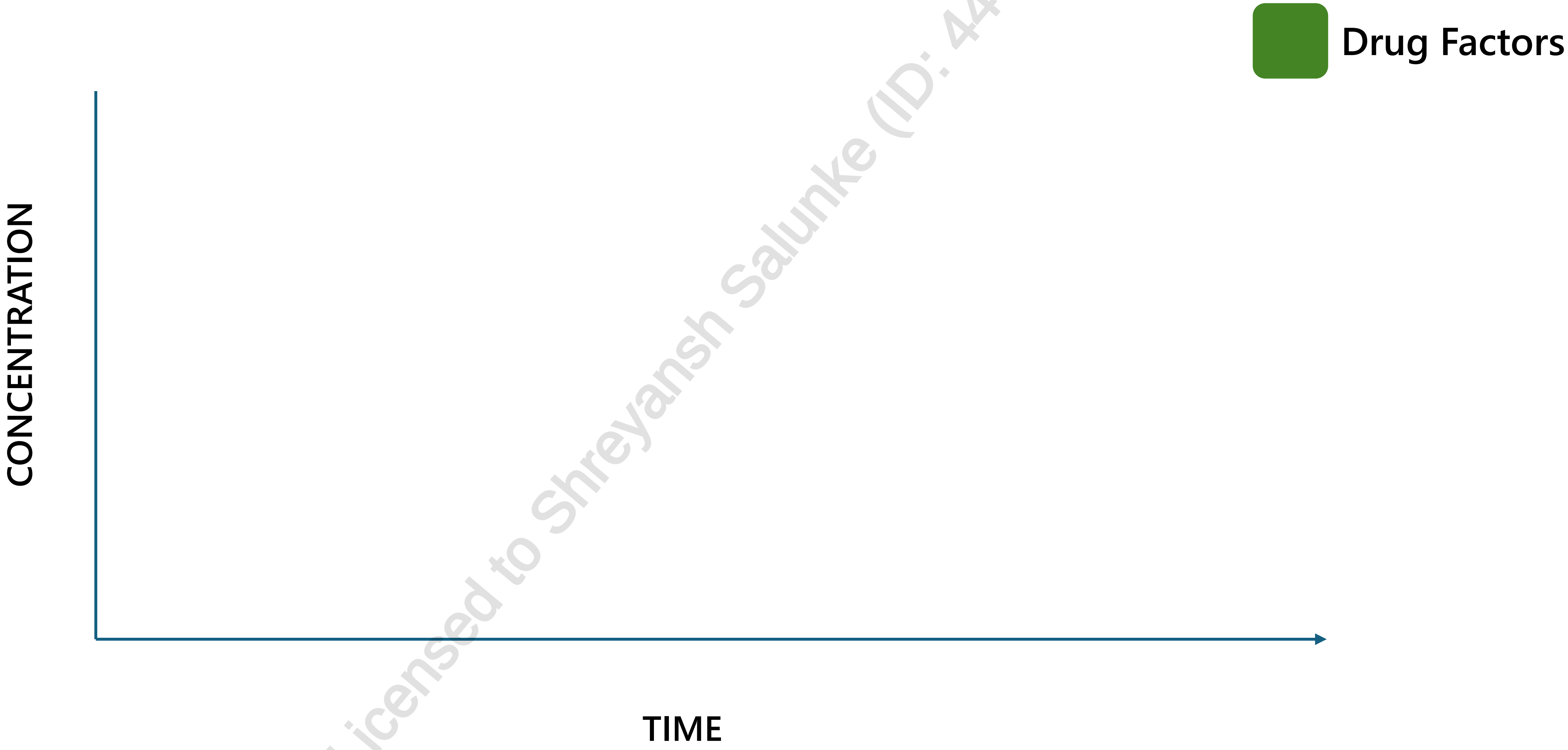

ANTIMICROBIAL DOSING AND ROUTE

Consider patient, syndrome and organism

Low Dose	High Dose
Localized infections Low infection burden Clinically stable patient Highly susceptible organism	Systemic infection High infection burden Unstable patients Moderately susceptible organism

Oral	Parenteral

Preferred
Requires:
    Hemodynamic stability
    Functioning and accessible GI tract
	Better for high dose (tolerance)
	Preferred in severe disease
	Possibly better spectrum of activity
	More costly

Drug Factors

Organism Factors

Host Factors

OPTIMIZING OUTCOMES AND TAILORING THERAPY

1. Gather and respond to microbiology results
2. Use a drug to which the organism is susceptible
3. Optimize drug for syndrome (right drug, dose, route and duration) to maximize effect → cure infection or prevent infection
4. Monitor patient response to infection
5. Optimize PK/PD to reduce risk of toxicity and increase efficacy
6. Minimize antibiotic exposure (ie. duration of therapy) to reduce resistance and adverse effects e.g. *C. difficile*-associated diarrhea

TAKEAWAYS

- Treatment of infectious diseases is a complex interplay between patient, microbiology and antimicrobial agents
- Understand the severity of illness, likely source of infection, pathogenesis, causative microbiology to consider antimicrobial therapy
- Host factors including comorbidity, age, weight, and clearance all impact choice and dose of antimicrobial agents
- Review antimicrobial spectrum of activity…extensively

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PRINCIPLES OF INFECTIOUS DISEASE & ANTIMICROBIAL THERAPY II

Qualifying Exam Preparatory Course

PRINCIPLES OF INFECTION

Case 2

A 72-year-old man (86 kg, 172 cm) presents with 3 days of fever, productive cough, and pleuritic chest pain.
Temp 38.9 °C, RR 24, BP 102/64, O_2 sat 90% RA. PMH: COPD, atrial fibrillation, stage 3 CKD (Cr 150 μmol/L, baseline 130), type 2 diabetes, mild cognitive impairment.

Medications: tiotropium, apixaban, metformin, atorvastatin, ramipril.
CXR: right lower-lobe consolidation with small effusion.
WBC 17 × 10^9/L, Na 133 mmol/L, glucose 14 mmol/L. Recent 10-day course of amoxicillin for dental abscess 2 weeks ago. No travel.
He lives with his wife who recently had influenza

AVAILABLE ANTIMICROBIAL AGENTS AND MECHANISMS OF ACTION

Drug Factors

DNA Synthesis Inhibitors

- Fluoroquinolones
- Nitroimidazoles (ex. Metronidazole)

Cell Wall Inhibitors

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- Lincosamides (ex. clindamycin)
- Oxazolidinones (ex. linezolid)

Folate Synthesis Inhibitors

- Sulfamethoxazole-Trimethoprim

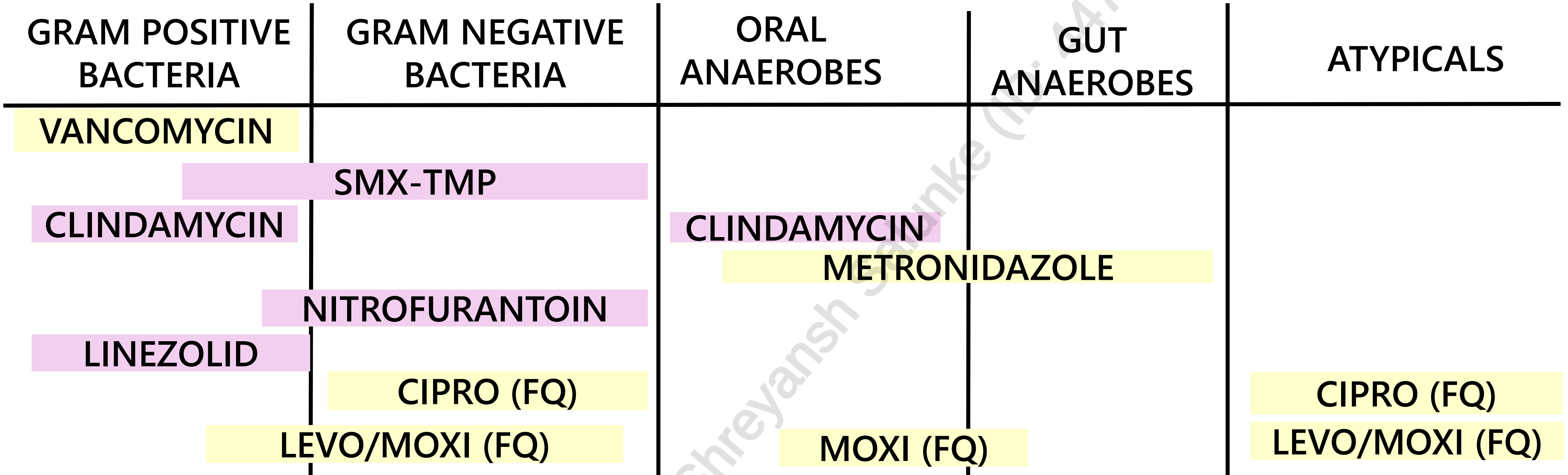
Miscellaneous

- Nitrofurantoin

SPECTRUM OF ACTIVITY OVERVIEW (simplified)

GRAM POSITIVE BACTERIA	GRAM NEGATIVE BACTERIA	ORAL ANAEROBES	GUT ANAEROBES	ATYPICALS
PENICILLIN		PENICILLIN		
AMOXICILLIN	AMOXICILLIN	AMOXICILLIN		
CLOXACILLIN				
PIP-TAZ	PIP-TAZ	PIP-TAZ	PIP-TAZ	
1ST GEN CEPHS	1ST GEN CEPHS			
2ND GEN CEPHS	2ND GEN CEPHS	1-4th GEN CEPHS		
3RD GEN CEPHS	3RD GEN CEPHS			
CARBAPENAMS	CARBAPENAMS	CARBAPENAMS	CARBAPENAMS	
MACROLIDES	MACROLIDES			MACROLIDES
TETRACYCLINES	TETRACYCLINES			TETRACYCLINES

SPECTRUM OF ACTIVITY OVERVIEW (simplified)



MSSA COVERAGE

PIP-TAZ or AMOX-CLAV
MOST CEPHALOSPORINS
(NOT CEFTAZIDIME)
CARBAPENAMS
LEVO/MOXI
CLOXACILLIN

MSSA & MRSA COVERAGE

VANCOMYCIN
LINEZOLID
SMX-TMP
CLINDAMYCIN
TETRACYCLINES
DAPTOMYCIN
TIGECYCLINE
CEFTOBIPROLE

BETA LACTAM COVERAGE OF GRAM POSITIVES

	Streptococcus pyogenes	*Staphylococcus aureus*	*Enterococcus faecalis*
Penicillin	Yes	No	Yes
Amoxicillin	Yes	No	Yes
Amoxicillin/clavulanate	Yes	Yes	Yes
Cloxacillin	Yes	Yes	No
Piperacillin/tazobactam	Yes	Yes	Yes
Cephalexein (1st gen)	Yes	Yes	No
Cefuroxime (2nd gen)	Yes	Yes (less than 1st gen)	No
Ceftriaxone (3rd gen)	Yes	Yes (less than 1st gen)	No
Ceftazidime (3rd gen)	No	No	No
Cefepime (4th gen)	Yes	Yes	No
Ceftolozane/tazobactam	No	No	No

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ANTIBIOGRAMS

QEII HSC Antibiotic Susceptibility (1 Jul 2020 – 30 Jun 2021)

Inpatients - Gram Negative Isolates: % Susceptible

	# tested**	Amoxicillin/Clavulanate	Ampicillin	Cefazolin (Cephalexin[1])	Ceftazidime	Ceftriaxone	Piperacillin/Tazobactam	Ertapenem	Meropenem	Ciprofloxacin	Fosfomycin[1]	Nitrofurantoin[1]	SXT/TMP	Doxycycline[2]	Gentamicin	Tobramycin
Enterobacteriales																
E. coli	876	80	54	79	92	91	95	99	100	73	100	96	80	76	93	92
Klebsiella pneumoniae	297	87	R	81	83	83	92	99	99	80		21	86	85	89	87
Klebsiella oxytoca	101	82	R	35	100	82	83	99	100	96	-	81	97	95	99	99
Citrobacter freundii	36	R	R	R	56*	53*	R	97	100	61	-	92	78	69	92	96
Enterobacter cloacae	171	R	R	R	63*	62*	R	87	99	94	-	38	93	91	98	100
Enterobacter aerogenes[4]	38	R	R	R	61*	58*	R	90	92	80	-	R	95	97	100	100
Serratia marcescens	86	R	R	R	99	89	-	97	100	76	-	R	100	17	98	97
Morganella morganii	70	R	R	R	86	94	99	100	100	93	-	R	94	R	94	100
Proteus mirabilis	179	100	89	91	100	97	99	100	100	96	-	R	88	R	94	100
Afermenters																
Pseudomonas aeruginosa	327	-	-	-	88	R	89	R	85	75	-	-	R	-	92	95
Burkholderia cepacia complex[3]	33	-	-	-	76	21	55	-	55	12	-	-	18	-	R	R
Stenotrophomonas sp.	45	-	-	-	58	R	R	-	R	22[5]	-	-	87	-	11	16
Acinetobacter baumannii complex[3]	49	-	-	-	81	-	83	-	93	88	-	-	91	-	93	92

All results are rounded to the nearest whole number.
Some isolates may have duplicates included which may alter the results.
* may develop resistance on treatment due to inducible AmpC B-lactamase (not recommended for treatment for Enterobacter and Citrobacter especially bacteremia/sterile sites other than uncomplicated UTI)
** approximate # tested as not all isolates tested against all antibiotics
"R" intrinsically resistant or susceptibility <10%

Susceptibility	<50%	50-89%	≥90%

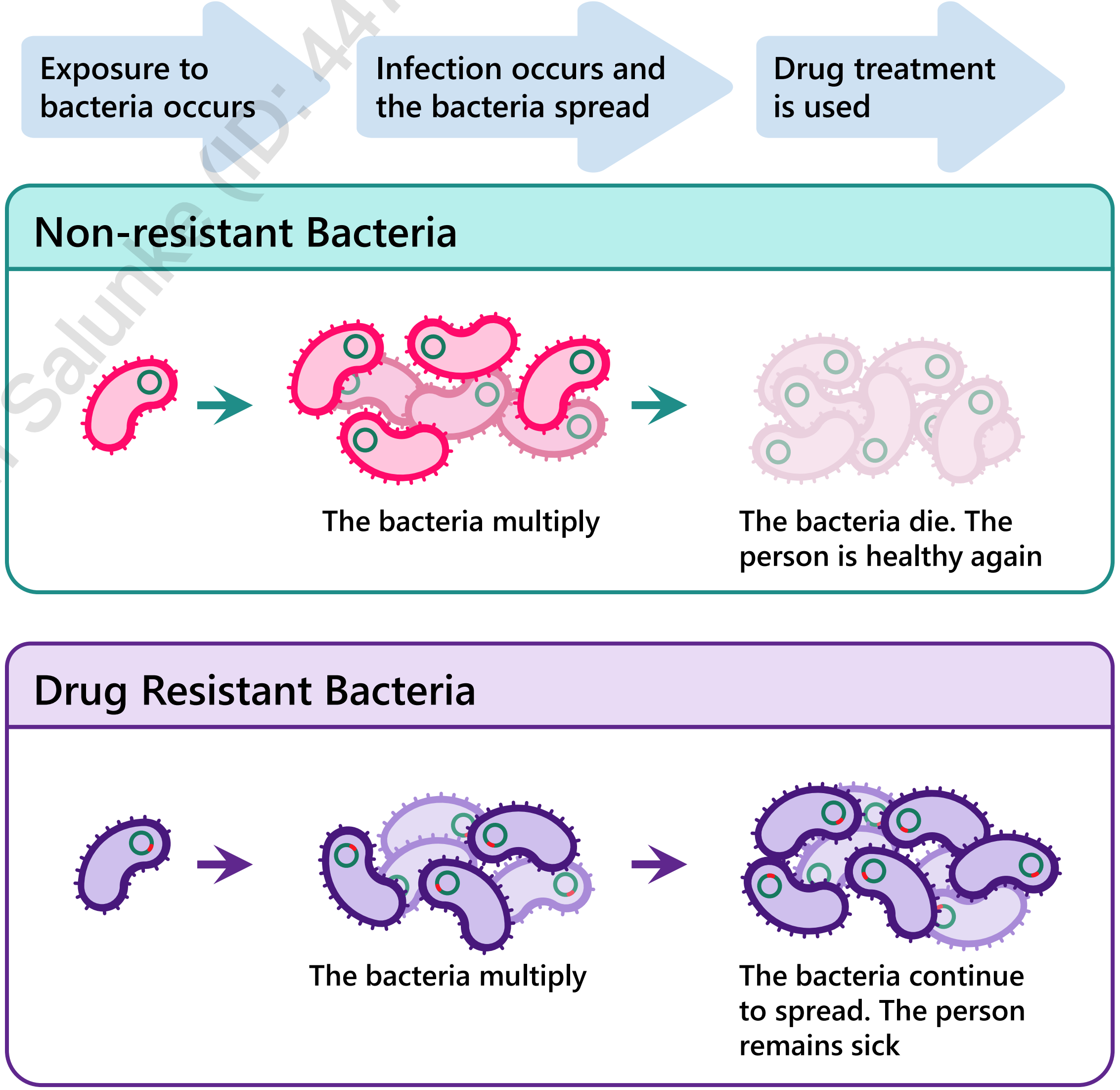
APPENDIX: WHAT'S NOT COVERED

* Amox/Amp does **NOT** cover *Klebsiella spp*

Atypical	*Gut Anaerobes*	*MSSA*	*MRSA*	*Entero-coccus*	*Group A/ S. pneumo*	*Enterobact/ Citrobact*	*H. influenzae*	*M. catarrhalis*	*PEcK*
Pens	Nope	Rare	Nope	Cloxacillin		Nope	Variable	Nope	Variable
Cephs	1^{st} /2^{nd} /3^{rd} /4^{th} gen ceph		1^{st} /2^{nd} /3^{rd} /4^{th} gen ceph	1^{st}/ 2^{nd} /3^{rd} /4^{th} gen ceph		1^{st}/2^{nd} gen ceph	1^{st} gen ceph	1^{st} gen ceph	
Carbapen			Nope						
BL/BLI			Nope	E.faecium		Nope			
Macrolides	Nope	Variable	Nope	Nope	Variable	Nope			Variable
FQs	Cipro/Levo (moxi variable)	Variable	Nope	Nope	Cipro				
Metro		Nope	Nope	Nope	Nope	Nope	Nope	Nope	Nope
Linez	Nope					Nope	Nope	Nope	Nope
Dapto	Nope					Nipe	Nope	Nope	Nope
AMG	Nope	Nope	Nope	Nope	Nope				
Vanco	Variable					Nope	Nope	Nope	Nope
Clinda	Variable				Variable	Nope	Nope	Nope	Nope
SMX/TMP	Nope			Variable	Variable				
Tetracycline	Nope			Variable		Variable			

PREVENTING RESISTANCE

1. Avoid inappropriate use of antibiotics
2. Avoid sub-therapeutic doses
3. Avoid prolonged duration of therapy
4. Vaccinate



ANTIMICROBIAL STEWARDSHIP

- Definition: *a coordinated series of interventions designed to optimize the use of antimicrobials in the management of infectious disease*
- Required operational practice for Hospital Accreditation Canada
- Antimicrobial stewardship consists of ID physician, pharmacists and perhaps microbiologist, lab technician, infection control practitioners, epidemiologist, etc.

PREGNANCY

"Safe Fetus MMeans Avoid These Now"

Antimicrobial	Clinical Pearls
Sulfamethoxazole/trimethoprim	• Avoid during 1^{st} and 3^{rd} trimesters • C/I near term
Fluoroquinolones*	• C/I in all 3 trimesters*
Macrolides Azithromycin Clarithromycin	• Safe • Generally, avoid
Metronidazole	• Avoid during 1^{st} trimester (note variable recommendations
Aminoglycosides*	• Avoid in 1^{st}, 2^{nd} and 3^{rd} trimesters
Tetracyclines	• C/I in all 3 trimesters
Nitrofurantoin	• C/I at term and during labour • 1^{st} trimester exposure controversial

BREASTFEEDING

Antimicrobial	Clinical Pearls
Metronidazole	• May stop breastfeeding for 12–24 hours to allow excretion of dose following a single dose of 2 g • Safe after the infant is 6 months old
Fluoroquinolones	• Use with caution (ideally avoid) – limited human data
Sulfamethoxazole/trimethoprim	• Safe in healthy term infants without G6PD deficiency • Proceed with caution if nursing infant has jaundice or is ill, stressed or premature
Nitrofurantoin	• Avoid when breastfeeding infants < 1 month old and G6PD deficiency

RENAL IMPAIRMENT

The following common antibiotics do not require dosage adjustment in patients with renal impairment:

Antibiotics	
• Azithromycin	• Linezolid
• Erythromycin	• Moxifloxacin
• Cloxacillin	• Fidaxomicin
• Penicillin VK	• Rifaximin
• Ceftriaxone	• Chloramphenicol
• Clindamycin	• Minocycline
• Doxycycline	• Tigecycline

BETA-LACTAM ALLERGIES

DEFINITION

***IMMUNOLOGIC HYPERSENSITIVITY REACTION** to an antibiotic in the beta-lactam class*

Classification	Symptoms	Time Frame
Type I – Immediate hypersensitivity (IgE)	Hives, angioedema, bronchospasm, hypotension, laryngeal edema, abdominal issues	Symptoms start *within 1 hour* of ingestion of *initial dose*
Type II Cytotoxic (IgG/IgM)	Thrombocytopenia, interstitial nephritis, hemolytic anemia	Within 72 hours, up to weeks
Type III Immune complex formation (complement)	Serum sickness syndrome	Within 1 – 3 weeks
Type IV Cell Mediated Hypersensitivity (T-cell)	Contact dermatitis, maculopapular eruptions, Stevens-Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN), drug rash with eosinophilia and systemic symptoms (DRESS)	Symptoms start after days of treatment

PRINCIPLES OF INFECTION

Quiz 1

Which antimicrobial provides *reliable coverage* for Pseudomonas aeruginosa?
A) Ceftriaxone
B) Piperacillin-tazobactam
C) Cefuroxime
D) Amoxicillin-clavulanate

PRINCIPLES OF INFECTION

Quiz 2

Which of the following agents has *definite* activity against MRSA?
A) Cefazolin
B) Linezolid
C) Amoxicillin
D) Ceftriaxone

PRINCIPLES OF INFECTION

Quiz 3

Which antimicrobial has *no reliable anaerobic coverage*?
A) Metronidazole
B) Piperacillin-tazobactam
C) Cefepime
D) Amoxicillin-clavulanate

PRINCIPLES OF INFECTION

Quiz 4

Which drug has *definite* coverage against Enterococcus faecalis, but not Enterococcus faecium?

A) Ampicillin
B) Cefazolin
C) Ciprofloxacin
D) Ceftriaxone

PRINCIPLES OF INFECTION

Quiz 5

Which antimicrobial provides *definite* coverage for ESBL-producing Enterobacterales?
A) Ertapenem
B) Piperacillin-tazobactam
C) Ceftriaxone
D) Amoxicillin-clavulanate

PRINCIPLES OF INFECTION

Quiz 6

Which antimicrobial has *definite* activity against atypical organisms (e.g., *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*)?
A) Cefuroxime
B) Azithromycin
C) Ceftriaxone
D) Amoxicillin

PRINCIPLES OF INFECTION

Quiz 7

Which of the following agents is most appropriate for treatment of *MSSA bacteremia*?
A) Cefazolin
B) Vancomycin
C) Clindamycin
D) Linezolid

PRINCIPLES OF INFECTION

Quiz 8

Which of the following antimicrobials has *definite coverage* for Pseudomonas aeruginosa but is less active against *Providencia* and *Morganella spp*.?

A) Ertapenem
B) Cefepime
C) Ceftolozane-tazobactam
D) Imipenem-cilastatin

Quiz 9

Which antibiotic provides *definite* coverage for anaerobes and Gram-positive cocci, but *no coverage for Gram-negatives*?

A) Clindamycin
B) Meropenem
C) Amoxicillin-clavulanate
D) Piperacillin-tazobactam

PRINCIPLES OF INFECTION

Quiz 10

Which antimicrobial provides definite coverage for *Pseudomonas*, anaerobes, *Staphyloccous aureus* and Enterococcus faecalis?
A) Ceftriaxone + Metronidazole
B) Piperacillin-tazobactam
C) Meropenem
D) Cefepime

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CHANGE LOG

June 2025

- Slide 4: re-defined virulence
- Added slides 26, 27
- Modified slides 33, 34
- Removed erythromycin from slide 40 (no longer marketed in Canada)

October 2025

- Slides 4, 17 & 25: Added case 1
- Slides 37 and 42: Added case 2
- Slides 51-60: Added Quiz questions

February 2026

- Slide 2: Updated NAPRA competencies
- Minor formatting changes